Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 50 epochs with a batch size of 32. The learning rate was set to 0.0003 with the Adam optimizer.

2 Results

Figure 1 shows the accuracy of DistilBERT by dataset.

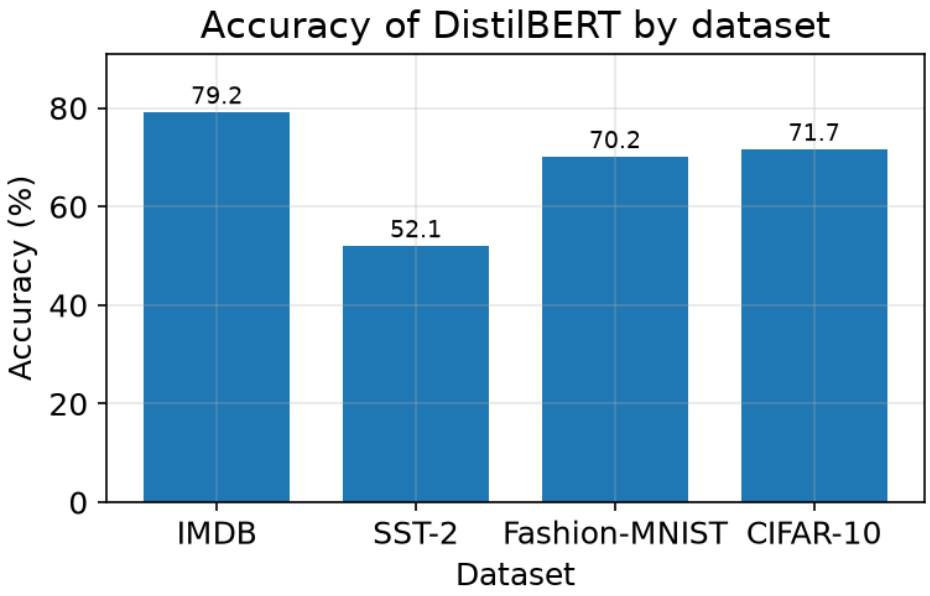


Figure 1: Accuracy of DistilBERT by dataset for the models in this experiment.

DistilBERT scores 27.1 points higher on IMDB than on SST-2 (Figure 1). As shown in Figure 1, on SST-2, DistilBERT obtains a accuracy of about 50%.

Figure 2 shows the training loss.

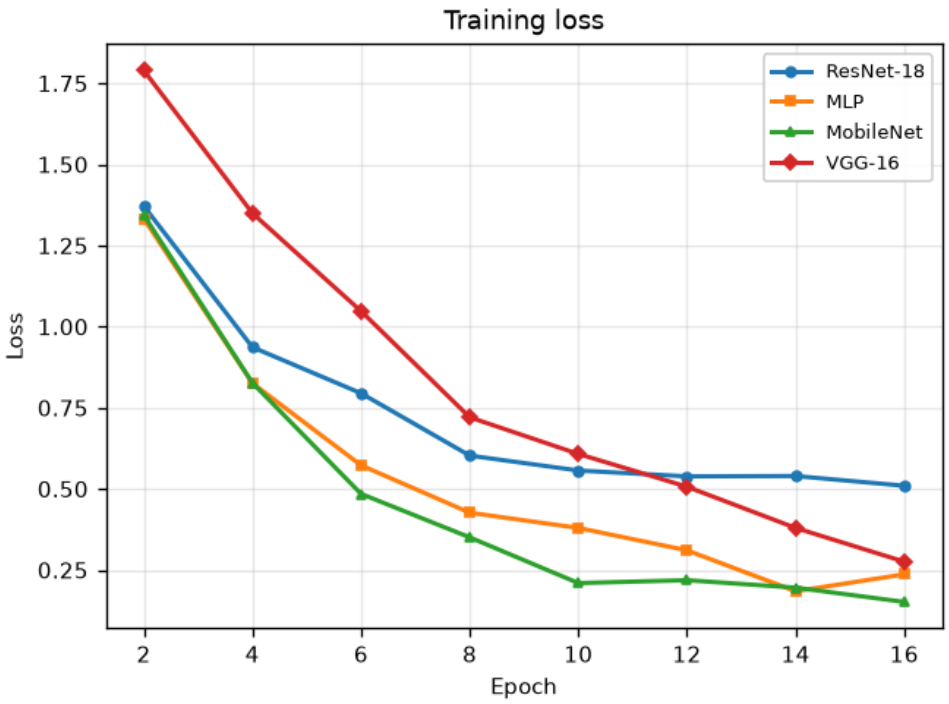


Figure 2: Training loss for the models in this experiment.

As shown in Figure 2, at epoch 10, the loss of ResNet-18 is 0.737. As shown in Figure 2, at epoch 2, the loss of VGG-16 is about 1.8.

Figure 3 shows the validation accuracy of BERT-base by dataset.

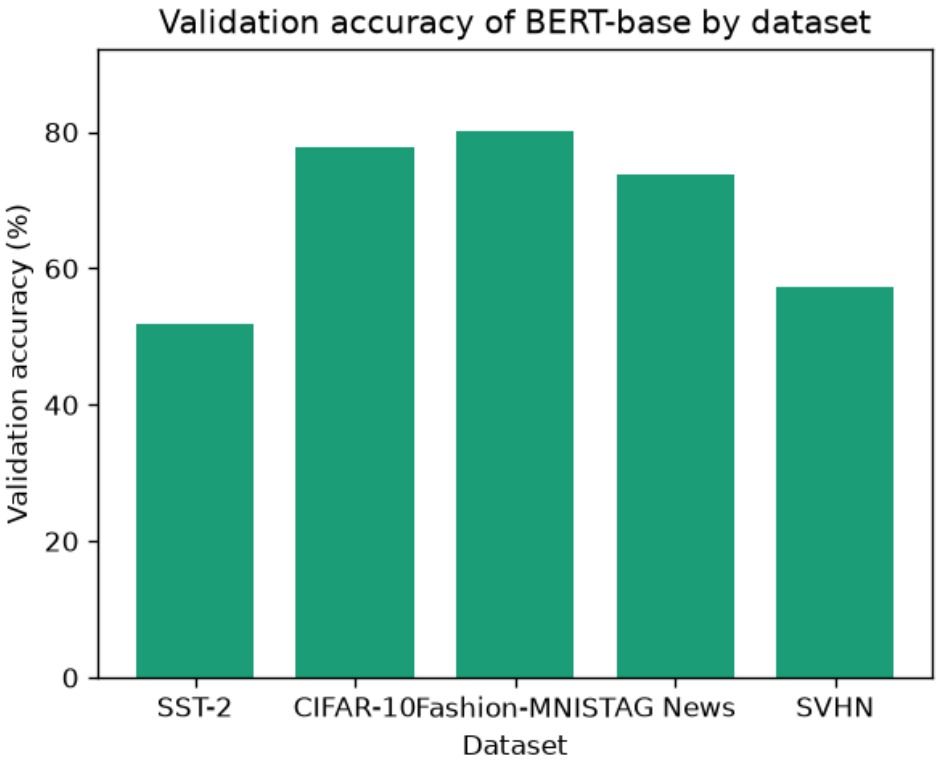


Figure 3: Validation accuracy of BERT-base by dataset for the models in this experiment.

On SVHN, BERT-base obtains a validation accuracy of 57% (Figure 3). On CIFAR-10, BERT-base obtains a validation accuracy of about 80%. BERT-base achieves about 80% validation accuracy on CIFAR-10.

Figure 4 shows the training loss.

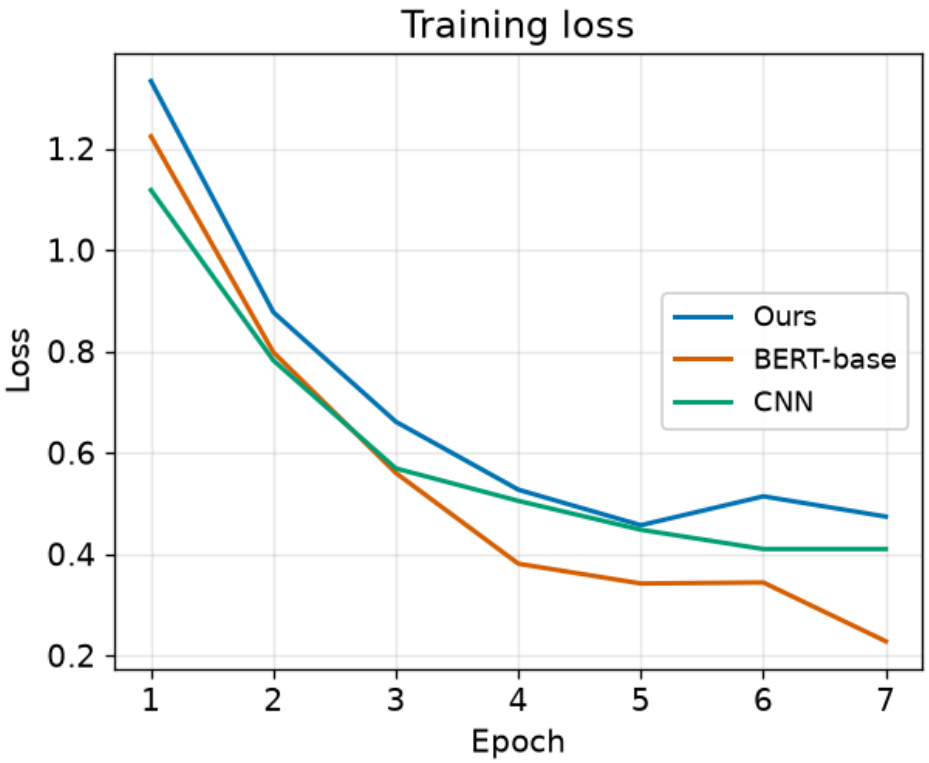


Figure 4: Training loss for the models in this experiment.

Ours has a loss of 1.211 at epoch 1. At epoch 3, the loss of BERT-base is 0.438.

3 Discussion

We note that the training was stable for all models.